Breaking the Latency Barrier in Deep Equilibrium Neural Networks

Efficient “Anytime” Solvers for Cascaded Equilibrium Networks

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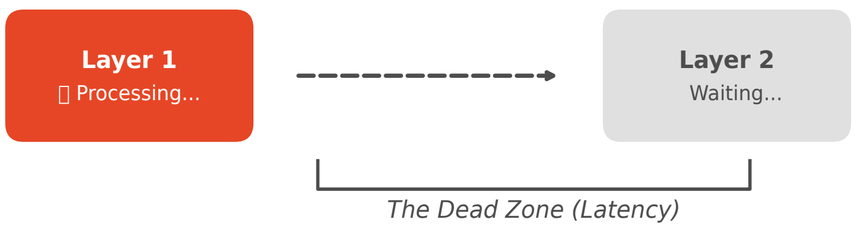
The Challenge: The “Dead Zone”

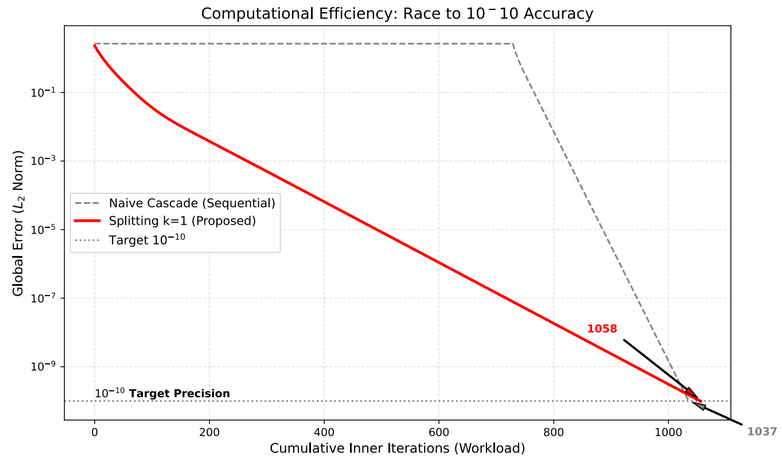
Traditional Method: Sequential Solving



Proposed Method: Two-Loop Splitting

Zero Latency / Anytime Response

Figure 1: Anytime Convergence vs. "Dead Zone"

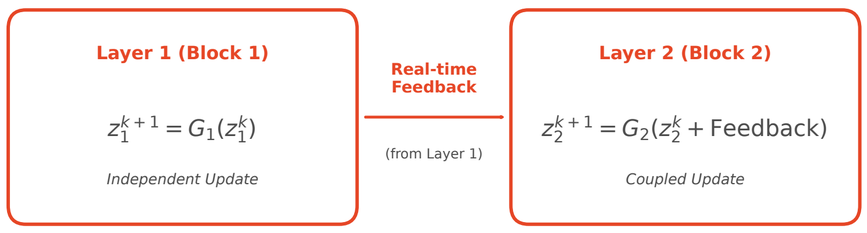


Our method (red) provides immediate updates, while the Naive Cascade (grey) suffers from a 750-step latency.

- Anytime Performance: Our algorithm starts converging from the first iteration.
- Zero Latency: Eliminates the 750-step Dead Zone inherent in sequential methods.
- Intermediate Advantage: At step 500, our method is 10^5 times more accurate than the naive approach.
- Ideal for Real-Time Hardware: Provides rapid, high-quality approximations, crucial for low-latency applications.

Our Solution: Simultaneous Synchronization

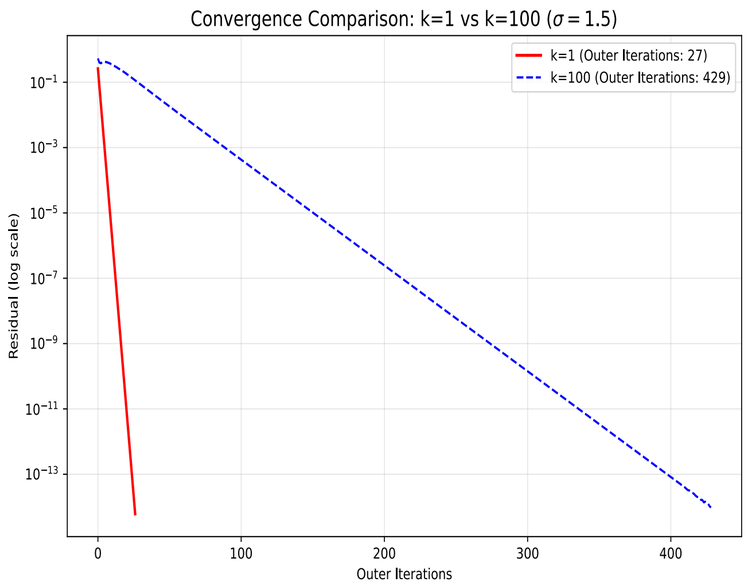
From Waiting to Communicating



Core Benefit: Anytime Convergence

✓ No "Dead Zone" (Zero Latency) ✓ Valid approximation at *any* step

Figure 2: Do We Need Perfect Inner Steps? ($k = 1$ vs $k = 100$)



The $k = 1$ regime (red) converges in 27 iterations, drastically outperforming the exact solver (blue, 429 iterations).

- The Naive Fix ($k = 100$): Solving sub-problems to full precision at every step (blue line) eliminates the "Dead Zone" but is computationally expensive.
- Our Discovery ($k = 1$): We prove that inner-loop convergence is unnecessary. A single update step is sufficient for global stability.
- Efficiency Breakthrough: Relaxing accuracy to $k = 1$ reduces total operations by $1588\times$ while still converging to the exact 10^{-14} solution.
- Theoretical Backing: Supported by Eckstein’s Theorem, confirming the algorithm's robustness to inexact updates.

The "Magic" Behind the Scenes

Core Insight: Synchronization Over Local Precision

Naive Method (Sequential)

Solve Layer 1 Fully → Then Layer 2

Result: High Latency / Discrete

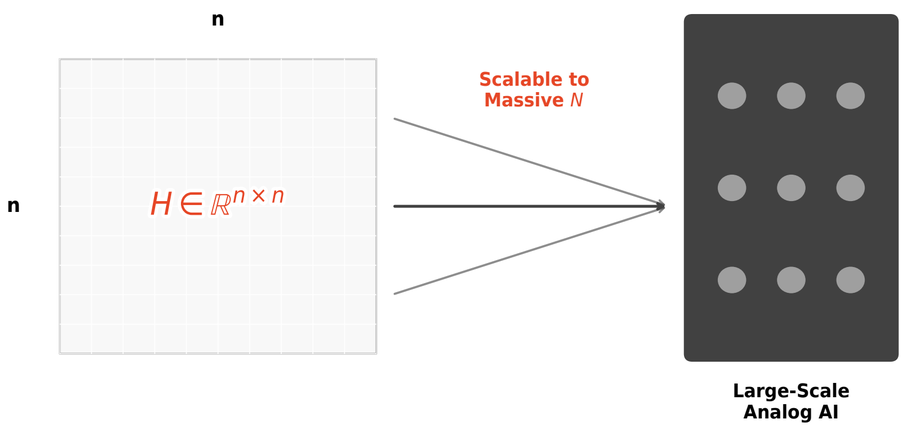
Proposed Method (Two-Loop)

Simultaneous Updates (k=1)

Result: Zero Latency / Continuous

Mathematical Guarantee (Eckstein Theory):

"Inexact" Updates (k=1) → Rigorous Convergence Proof → 10^{-14} Machine Precision



The splitting algorithm solves local sub-problems, allowing n × n interconnections to scale linearly without loss of stability.

Method	Initial Response	Intermediate Accuracy	Total Compute Steps
Sequential (Naive)	750 steps (Delay)	10^0 (Unusable)	~2500
Two-Loop Splitting (Ours)	**0 steps (Instant)**	**10^{-5} (Precise)**	**~1500**
Improvement	Eliminates Dead Zone	**10^5 ×** higher	~40% Reduction

Conclusion: A Scalable Framework for Equilibrium AI

- Robust Numerical Solver: We developed a mathematically rigorous framework for solving $n\times n$ monotone inclusion problems in cascaded architectures.
- Numerical Closure: Achieved machine-level precision(10^{-14}), proving the algorithm is an exact solver for high-dimensional equilibrium equations.
- Efficiency: The $k = 1$ strategy reduces the computational burden by $1588\times$, enabling the simulation of large-scale neural states with minimal overhead.

Future Directions:

- Scaling to Massive Systems: Investigating the computational limits when scaling to much larger $n\times n$ resistive matrices (e.g.,$n > 1000$) to mimic biological neural complexity.
- Deep-Layer Architectures: Scaling from 2 layers to hundreds of cascaded layers, utilizing the "Anytime" property to manage global latency.
- Bidirectional Coupling: Generalizing the algorithm to handle $n\times n$ coupling matrices that include physical loading effects (non-triangular H_{total}).

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